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24CS320

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

III Semester B.E Degree Examination

Department of Computer Science and Engineering

DIGITAL SYSTEM DESIGN

Duration: 3 Hours | Max. Marks:100

Instructions:

- PART- A Questions are Compulsory.
- PART- B Questions are having internal choice.

PART – A

Q.NO	CO	CD	PI	MARKS
1.	CO1	L3	1.3.1	4+6
	a. Simplify, $F = (X+ Y)\ (X'(Y' + Z'))' + X'(Y + Z)$ using Boolean laws and theorems. b. Write a note on conversion between SOP and POS considering a three variable truth table.			
2.	CO2	L3	1.4.1	10
	Show how 4 variable truth table can be realized using an 8-to-1 multiplexer.			
3.	CO3	L2	2.3.2	10
	Explain the working of NOR-Gate Latch and NAND-Gate Latch with logic symbol and truth table.			
4.	CO4	L3	2.2.3	10
	Explain the working of Ring counter and Switched Tail counter with diagrams.			
5.	CO5	L3	2.3.2	10
	Design a Mod-8 Synchronous Counter using J-K flipflops.			

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PART – B

Q.NO	CO	CD	PI	MARKS
6.	CO1	L3	1.4.1	10
	Design a system which accepts four inputs a,b,c,d and outputs a 1 whenever the least significant input or the most significant input goes high. Use Karnaugh map to simplify the expression and design the circuit.			
	OR			
7.	CO1	L3	2.2.3	10
	For the following Boolean function, use the Quine-McClusky method to obtain all the Prime Implicants. $F(a,b,c,d) = \Sigma m(7,9,12,13,14,15)+dc(4,11)$.			
8.	CO2	L2	1.4.1	10
	With logic diagram and truth table explain BCD to Decimal decoder.			
	OR			
9.	CO2	L2	1.3.1	10
	With required diagrams, describe parity generators and checkers.			
10.	CO3	L2	2.2.3	10
	Explain JK Master Slave flipflop with required diagrams.			
	OR			
11.	CO3	L2	2.2.4	10
	Write a note on conversion of flip flops.			
12.	CO4	L2	2.2.4	10
	Discuss in detail about Shift registers types with diagrams.			
	OR			
13.	CO4	L2	2.2.4	5+5
	Discuss with logic diagram (i) Serial in Serial Out (SISO) (ii) Parallel in and Parallel Out (PIPO) registers.			

14.	CO5	L2	2.3.2	10
	What are Asynchronous Counters? Explain three bits asynchronous down counter with logic circuit, truth table and waveforms.			
	OR			
15.	CO5	L2	2.2.4	10
	Explain two distinct models by which a synchronous sequential logic circuit can be designed with state transition diagrams and state synthesis table.			

Course Outcome (CO):	
CO–1	Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions.
CO–2	Design and implement the combinational circuits.
CO–3	Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops.
CO–4	Illustrate the working of various registers and register circuits.
CO–5	Design and implement synchronous/asynchronous counters and synchronous sequential circuits.

PI	
1.3.1	*Apply Engineering Fundamentals.*
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions.
2.2.4	Compare and contrast alternative solution/methods to select the best methods.
2.3.2	Identify design constraints for required performance criteria.

CD		
L1 - Remember	L2 - Understand	L3 - Application

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